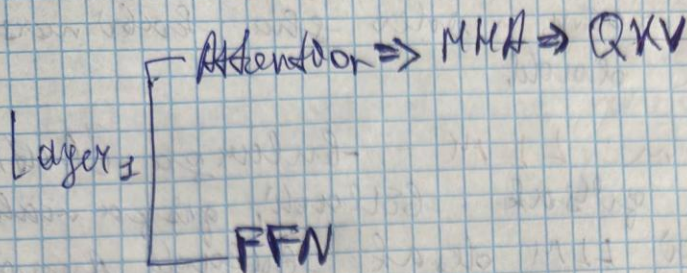


LLM 2-oy 30-Days Report.



Training - bu modelning parametrlarini
organizatsiya qilishning boshqarish
larini laboriya yaxshilash jarayoni.

Input $\Rightarrow$ Tokenization $\Rightarrow$ Token ID $\Rightarrow$
Vocab $\Rightarrow$ Embedding $\Rightarrow$ Layer $\Rightarrow$ Attention $\Rightarrow$ MHA $\Rightarrow$ Training
FFN

Model parametrlari

Data / Data Preprocessing

Tokenizer

Embeddings

Transformer blocks

Attention

MLP

LM Head

Building the model $\neq$ Training the model.

Model Building

BE
fishab soli

Bend
musol!

GPT
musol!

LLM

Sel!

osh

Model

loss
pred

Train

$\Rightarrow$ G

Org
rate

yan

en

ja

Pretraining Transformers

BERT

hushab golshad talabotadi

GPT

sharoq olabon sharoq

Bert
Model

! Osh Poytaxti

Outputs Tokent

GPT
Model

the cost is

Outputs sleeping - 0.5
eating - 0.25
walking - 0.11

LLM da Dataset buildolarni ichida
Self Supervised Learning amalga
oshadi.

LOSS

Model qanchalik xato qilayotgand.
Loss tells the model how bad its current
prediction is.

The Training Loop

Training data $\Rightarrow$ Loss $\Rightarrow$ Optimizer
 $\Rightarrow$ Gradients.

Learning Rate

Reganish qadamlar. Learning
rate scheduling bilan jarayon
yanada yaxshi lanadi. his xam
emas kam xam emas yaxshi
jarayon maqul.

- ALL
- ① Training LLM Fundamentals
 - ② Loss & Backpropagation
 - ③ Optimizer & Adam W
 - ④ Learning Rate scheduling
 - ⑤ Training

configs $\Rightarrow$ run.py, yaml - ichida training konfiguratsiya beriladi.

src. seed.py - buni ichida faqatgina poyga boshlashni ta'minlayotgan, boshlanish nuqtasini belgilaydi. Chiqishni shartlarni amalga oshirishni aylanib o'tib beradi.

src. lr_scheduler - learning rate ni to'g'ri tartibda o'zgartirish, yaxshiroq ta'minlash. LLM ni weightlarni bosh b-n to'g'ri tanlash shart.

src. model.py - configure optimizers. Adam W optimizatsiya qilishni ta'minlaydi. Yana boshqa optimizatsiya qilish.

src. checkpoint.p - faqat xalqlar un training process ni ta'minlashni ta'minlaydi.

src. train_config.py - reads some python objects can we training ni ta'minlash bo'lgan shartlar.

src. trainer.py - boshlanish training ni ta'minlash. Yana boshqa optimizatsiya qilish.

scripts. train.py - the command you run to start a training run.